

Math Thermo
class # 22
4/14/2026

Recapitulation of Equations

$$(22.1) \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0,$$

$$(22.2) \quad \frac{\partial (\rho \phi_i)}{\partial t} + \nabla \cdot (\rho \phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k,$$

$$(22.3) \quad \frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v} + \Pi) = -\nabla P - \sum_{i=1}^r \rho_i \nabla \psi_i,$$

$$(22.4) \quad \frac{\partial}{\partial t} (\rho u) + \nabla \cdot (\rho u \vec{v}) = -\nabla \cdot \vec{J}_u^D - P \nabla \cdot \vec{v} \\ - \Pi : \nabla \vec{v} - \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D,$$

$$(22.5) \quad \partial_t (\rho s) + \nabla \cdot (\rho s \vec{v}) = -\nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$(22.6) \quad \vec{J}_s^D := \frac{1}{T} \left(\vec{J}_u^D - \sum_{i=1}^r \mu_i \vec{J}_i^D \right),$$

$$\sigma_s := \vec{J}_u^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T}$$

$$(22.7) \quad - \frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \left(T \nabla \left(\frac{\mu_i}{T} \right) + \nabla \psi_i \right) \\ - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k.$$

The forms of $\vec{J}_u^D$, Π , $\vec{J}_i^D$, A_k must be chosen so that $\sigma_s \geq 0$ pointwise. Furthermore an equation of state for the hydrostatic pressure must be chosen.

An Alternate Form for σ_s

let us use the identity

$$\begin{aligned} T \nabla \left(\frac{\mu_i}{T} \right) &= \frac{T}{T} \nabla \mu_i + T \mu_i \nabla \left(\frac{1}{T} \right) \\ &= \nabla \mu_i + T \mu_i \nabla \left(\frac{1}{T} \right). \end{aligned}$$

Then,

$$\begin{aligned} \sigma_s &= \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T} \\ &\quad - \frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \left(\nabla \mu_i + T \mu_i \nabla \left(\frac{1}{T} \right) + \nabla \psi_i \right) \\ &\quad - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k \\ &= \left(\vec{J}_v^D - \sum_{i=1}^r \mu_i \vec{J}_i^D \right) \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T} \\ &\quad - \frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \left(\nabla \mu_i + \nabla \psi_i \right) \\ &\quad - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k \end{aligned}$$

Therefore we may write

$$\begin{aligned} \sigma_s &= T \vec{J}_s^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\Pi : \nabla \vec{v}}{T} \\ (22.8) \quad &\quad - \frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \nabla \mu_i^e - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\sigma}_k, \end{aligned}$$

where

$$\mu_i^e := \mu_i + \psi_i$$

is the extended chemical potential.

Finally, recall that

$$\sum_{i=1}^r \vec{J}_i^D = \vec{0}$$

Therefore

$$(22.9) \quad -\frac{1}{T} \sum_{i=1}^r \vec{J}_i^D \cdot \nabla \mu_i^e = 0.$$

Subtracting (22.9) from (22.8) we have

$$(22.10) \quad \sigma_s = T \vec{J}_s^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\pi : \nabla \vec{v}}{T} \\ - \frac{1}{T} \sum_{i=2}^r \vec{J}_i^D \cdot \nabla \tilde{\mu}_i^e - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\mathcal{F}}_k,$$

where

$$\tilde{\mu}_i^e := \mu_i^e - \mu_1^e \\ = \mu_i + \psi_i - \mu_1 - \psi_1, \quad i = 2, \dots, r,$$

is called the exchange chemical potential.

There are, of course, lots of equivalent forms...

The Cauchy Stress Tensor

We have previously seen the utility of the decomposition

$$\mathcal{S} = P\mathbf{I}_3 + \mathbb{T},$$

where P is the thermodynamic, hydrostatic pressure. To further analyze $\mathcal{S}$, let us recall the following generic decomposition.

Proposition (22.1): For any $A \in \mathbb{R}^{n \times n}$, the decomposition

$$A = a\mathbf{I}_n + \mathring{A}_s + A_A$$

holds, where $\mathring{A}_s \in \mathbb{R}^{n \times n}$ is deviatoric, i.e.,

$$(\mathring{A}_s)^T = \mathring{A}_s \text{ and } \text{tr}(\mathring{A}_s) = 0,$$

and $A_A \in \mathbb{R}^{n \times n}$ is anti-symmetric, i.e.,

$$(A_A)^T = -A_A.$$

Proof: Observe that

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

Set

$$A_A := \frac{1}{2}(A - A^T).$$

Clearly, $(A_A)^T = -A_A$, i.e. A_A is anti-symmetric.

Now, observe that

$$\frac{1}{2}(A+A^T) = \frac{1}{n}\text{tr}(A)I_n$$

$$+ \frac{1}{2}(A+A^T) - \frac{1}{n}\text{tr}(A)I_n.$$

Set

$$a := \frac{1}{n}\text{tr}(A)$$

$$\mathring{A}_s := \frac{1}{2}(A+A^T) - \frac{1}{n}\text{tr}(A)I_n.$$

Clearly, $(\mathring{A}_s)^T = \mathring{A}_s$. Finally,

$$\text{tr}(\mathring{A}_s) = \sum_{i=1}^n \frac{1}{2}(a_{ii} + a_{ii}) - \frac{1}{n}\text{tr}(A) \sum_{i=1}^n \delta_{ii}$$

$$= \frac{1}{2} \cdot 2 \text{tr}(A) - \frac{1}{n}\text{tr}(A) \cdot n$$

$$= 0. \quad ///$$

From the decomposition

$$A = P I_3 + \Pi,$$

let us set (using the last proposition)

$$\Pi = p I_3 + \mathring{\Pi}$$

where $\mathring{\Pi}$ is deviatoric, i.e.

$$\mathring{\Pi}^T = \mathring{\Pi} \quad \text{and} \quad \text{tr}(\mathring{\Pi}) = 0,$$

and $p \in \mathbb{R}$ is the non-hydrostatic pressure.

Observe that, since Π is symmetric (S is symmetric) there is no anti-symmetric part in the decomposition.

Next, we wish to apply Proposition (22.1) to the non-symmetric tensor $\nabla \vec{v}$:

$$\nabla \vec{v} = \frac{1}{3} \nabla \cdot \vec{v} \mathbf{I}_3 + \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T) - \frac{1}{3} \nabla \cdot \vec{v} \mathbf{I}_3 + \frac{1}{2} (\nabla \vec{v} - (\nabla \vec{v})^T).$$

The deviatoric tensor

$$\tau := \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T) - \frac{1}{3} \nabla \cdot \vec{v} \mathbf{I}_3$$

is well known in Newtonian fluid mechanics as the shear rate tensor. Thus

$$\nabla \vec{v} = \frac{1}{3} \nabla \cdot \vec{v} \mathbf{I}_3 + \tau + \frac{1}{2} (\nabla \vec{v} - (\nabla \vec{v})^T).$$

Proposition (22.2): The term $\Pi : \nabla \vec{v}$, which appears in the entropy production term σ_s , can be expressed as

$$\begin{aligned} \nabla \vec{v} : \Pi &= \left(\frac{1}{3} \nabla \cdot \vec{v} \mathbf{I}_3 + \tau + \frac{1}{2} (\nabla \vec{v} - (\nabla \vec{v})^T) \right) : (p \mathbf{I}_3 + \overset{\circ}{\Pi}) \\ (22.11) \quad &= \frac{p}{3} \nabla \cdot \vec{v} + \tau : \overset{\circ}{\Pi} \end{aligned}$$

Proof: A simple exercise. ///

Thus, we have the equivalent expressions for the entropy production rates:

$$\begin{aligned}
 \sigma_s := & \vec{J}_v^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\rho \nabla \cdot \vec{v}}{3T} - \frac{\gamma : \dot{\Pi}}{T} \\
 & - \frac{1}{T} \sum_{i=1}^5 \vec{J}_i^D \cdot \left(T \nabla \left(\frac{\mu_i}{T} \right) + \nabla \psi_i \right) \\
 & - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\mathcal{F}}_k,
 \end{aligned}
 \tag{22.12}$$

$$\begin{aligned}
 \sigma_s = & T \vec{J}_s^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\rho \nabla \cdot \vec{v}}{3T} - \frac{\gamma : \dot{\Pi}}{T} \\
 & - \frac{1}{T} \sum_{i=2}^r \vec{J}_i^D \cdot \nabla \tilde{\mu}_i^e - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\mathcal{F}}_k.
 \end{aligned}
 \tag{22.13}$$

Variational Derivatives

Suppose that

$$F[\phi] = \int_{\Omega} f(\phi, \nabla \phi, \Delta \phi) d\vec{x} \quad \Omega \subseteq \mathbb{R}^d$$

where

$$f: \mathbb{R} \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$$

is a smooth function. Suppose that

$$\phi: \Omega \times [0, T] \rightarrow \mathbb{R},$$

that is, ϕ is a function of space and time. Then F is a function of time for a given function ϕ .

To make the calculations tractable, let us write

$$\overset{f}{(\phi, \vec{q}, r)} \mapsto f(\phi, \vec{q}, r) \in \mathbb{R}$$

Thus, $\frac{\partial f}{\partial \phi}$ is the derivative with respect to the first variable, $\frac{\partial f}{\partial q_i}$ is derivative with respect to the $i+1^{\text{st}}$ variable and $\frac{\partial f}{\partial r}$ is the derivative with respect to the $d+2^{\text{nd}}$ variable.

let us take the time derivative

$$\begin{aligned} \dot{F} &= \frac{d}{dt} F[\phi](t) \\ &= \int_{\Omega} \left\{ \frac{\partial f}{\partial \phi} \dot{\phi} + \sum_{i=1}^d \frac{\partial f}{\partial q_i} \left(\frac{\partial \dot{\phi}}{\partial x_i} \right) + \frac{\partial f}{\partial r} (\Delta \dot{\phi}) \right\} d\vec{x} \end{aligned} \quad \begin{array}{l} \text{chain rule} \\ \swarrow \end{array}$$

where

$$\dot{q} := \frac{\partial q}{\partial t}.$$

let us use integration - by - parts:

$$\int_{\Omega} g \frac{\partial f}{\partial x_i} d\vec{x} = - \int_{\Omega} \frac{\partial g}{\partial x_i} f d\vec{x} + \int_{\partial\Omega} g f n_i dS$$

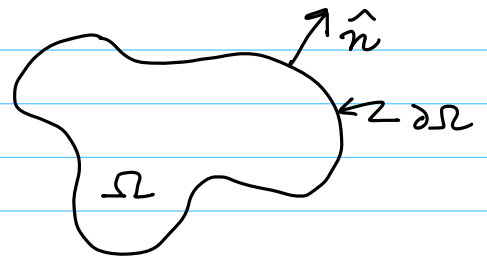
$$\int_{\Omega} g \Delta f d\vec{x} = \int_{\Omega} \Delta g f d\vec{x} + \int_{\partial\Omega} \left\{ g \frac{\partial f}{\partial n} - f \frac{\partial g}{\partial n} \right\} dS$$

where

$$\frac{\partial f}{\partial n} = \nabla f \cdot \hat{n}$$

and

$$[\hat{n}]_i = n_i$$



is the outward - pointing unit normal on $\partial\Omega$.
Assuming that ϕ is smooth enough so that

$$\left(\frac{\partial \dot{\phi}}{\partial x_i} \right) = \frac{\partial}{\partial x_i} (\dot{\phi})$$

and

$$(\Delta \dot{\phi}) = \Delta (\dot{\phi})$$

it follows that

$$\begin{aligned} \dot{F} &= \int_{\Omega} \left\{ \frac{\partial f}{\partial p} \dot{\phi} + \underbrace{\frac{\partial f}{\partial q}}_{\sum_{i=1}^d \frac{\partial f}{\partial q_i} \frac{\partial \dot{\phi}}{\partial x_i}} \cdot \nabla \dot{\phi} + \frac{\partial f}{\partial r} \Delta \dot{\phi} \right\} d\vec{x} \\ &= \int_{\Omega} \left\{ \frac{\partial f}{\partial p} - \nabla \cdot \frac{\partial f}{\partial \vec{q}} + \Delta \frac{\partial f}{\partial r} \right\} \dot{\phi} d\vec{x} \\ &\quad + \int_{\partial\Omega} \frac{\partial f}{\partial \vec{q}} \cdot \hat{n} \dot{\phi} dS + \int_{\partial\Omega} \left\{ \frac{\partial f}{\partial r} \frac{\partial \dot{\phi}}{\partial n} - \dot{\phi} \frac{\partial}{\partial n} \left(\frac{\partial f}{\partial r} \right) \right\} dS. \end{aligned}$$

We will assume that ϕ is from a class of functions such that the boundary integrals vanish. If this is the case,

$$\dot{F} = \int_{\Omega} \frac{\delta F}{\delta \phi} \dot{\phi} d\vec{x},$$

where

$$(21.16) \quad \frac{\delta F}{\delta \phi} = \frac{\partial f}{\partial \phi} - \nabla \cdot \frac{\partial f}{\partial \vec{q}} + \Delta \frac{\partial f}{\partial r}$$

is called the variational derivative of F .

Example (21.1): Suppose that

$$F[\phi] = \int_{\Omega} \left\{ f_0(\phi) + \frac{\kappa}{2} |\nabla \phi|^2 \right\} d\vec{x}.$$

Thus,

$$f(\phi, \vec{q}) = f_0(\phi) + \frac{\kappa}{2} \vec{q} \cdot \vec{q}.$$

Assume that, for each $t \in [0, T]$,

$$\phi(t) \in \mathcal{A} = \left\{ \psi \in C^2(\bar{\Omega}) \mid \frac{\partial \psi}{\partial n} = 0 \right\}.$$

Then

$$\begin{aligned} \dot{F} &= \int_{\Omega} \left\{ f'_0(\phi) \dot{\phi} + \kappa \nabla \phi \cdot \nabla \dot{\phi} \right\} d\vec{x} \\ &= \int_{\Omega} \left\{ f'_0(\phi) - \nabla \cdot (\kappa \nabla \phi) \right\} \dot{\phi} d\vec{x} \\ &\quad + \int_{\partial \Omega} \kappa \frac{\partial \phi}{\partial n} \dot{\phi} dS \end{aligned}$$

Thus,

$$\frac{\delta F}{\delta \phi} = f'_0(\phi) - \nabla \cdot (\kappa \nabla \phi).$$

Application : Two-Phase Flow

We will assume that

$$1) \quad r = 2, \text{ (binary mixture)}$$

$$2) \quad \Psi_i \equiv 0, \quad i = 1, 2, \quad (\text{no external potential})$$

$$3) \quad \sigma_k \equiv 0, \quad k = 1, \dots, K, \quad (\text{no reactions})$$

$$4) \quad p \equiv p_0 \equiv \text{constant} \quad (\text{incompressible})$$

Since the mixture is incompressible, the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

is replaced by

$$\nabla \cdot \vec{v} = 0$$

The partial densities are

$$\rho_i = \rho_0 \phi_i, \quad i = 1, 2.$$

where

$$\phi_1 + \phi_2 = 1$$

are the mass fractions.

We will assume that

$$U = \int_{\Omega} \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

$$F = \int_{\Omega} \hat{f}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

$$S = \int_{\Omega} \hat{s}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

To shorten the writing, we will use the notation

$$\hat{u}(T, \phi, \nabla \phi) = \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2).$$

We will assume (more on this later) that

$$\hat{f} = \hat{u} - T \hat{s}$$

as in equilibrium. This is another consequence of local thermodynamic equilibrium.

Further, we assume, for simplicity,

$$\hat{u}(T, \phi, \nabla \phi) = u(T, \phi) + \lambda_u \delta(\phi, \nabla \phi),$$

$$\hat{f}(T, \phi, \nabla \phi) = f(T, \phi) + \lambda_f \delta(\phi, \nabla \phi),$$

$$\hat{s}(T, \phi, \nabla \phi) = s(T, \phi) + \lambda_s \delta(\phi, \nabla \phi).$$

In other words, the T and $\nabla \phi$ dependencies are separable. For u, f, s we take

$$u = p_0 C_n T$$

$$f = p_0 C_n T - p_0 C_n T \ln\left(\frac{T}{T_0}\right)$$

$$S = p_0 C_h \ln\left(\frac{T}{T_0}\right)$$

Clearly,

$$f = u - Ts$$

For the gradient term, we use the model

$$\delta(\phi, \nabla\phi) = \frac{1}{\varepsilon} \frac{\phi_1^2 \phi_2^2}{4} + \frac{\varepsilon}{2} \left(\frac{|\nabla\phi_1|^2}{2} + \frac{|\nabla\phi_2|^2}{2} \right),$$

where $\varepsilon \gamma_0$ has units of length.

Finally, we note that, it must be true that

$$\lambda_f = \lambda_u - T \lambda_s.$$

Now, let us note that

$$\frac{\partial s}{\partial T} = p_0 C_h \frac{1}{T}$$

$$\frac{\partial u}{\partial T} = p_0 C_h$$

so that

$$\frac{\partial s}{\partial T} \left(\frac{\partial u}{\partial T} \right)^{-1} = \frac{1}{T}$$

This condition is called thermal compatibility.